5.2 Network security	**5.2.1** Understand the importance of network security and be able to use appropriate validation and authentication techniques (access control, physical security and firewalls). **5.2.2** Understand security issues associated with the 'cloud' and other contemporary storage. **5.2.3** Understand different forms of cyber attack (based on technical weaknesses and behaviour), including social engineering (phishing, shoulder surfing, pharming), unpatched software, USB devices, digital devices and eavesdropping. **5.2.4** Understand methods of identifying vulnerabilities, including penetration testing, ethical hacking, commercial analysis tools and review of network and user policies. **5.2.5** Understand how to protect software systems from cyber attacks, including considerations at the software (application) design stage, audit trails, securing operating systems, secure coding, code reviews to remove code vulnerabilities in programming languages and bad programming practices, modular testing and effective network security provision.